

Syllabus

Fall Semester, 2025

Course Title	Course Code	Type	Credits	Hours
Elementary Real Analysis	MC4102	Advanced Elective	3	3

Instructor	Section	Office	Office Hours	Phone	Email	Classroom
Jongmin Kim	1	H3507	Mon7 Wed3 Thu3	051-606-2149	kimjongmin@ksa.hs.kr	H3303

1. Course Description

This course offers a rigorous introduction to real analysis, focusing on the theoretical foundations of calculus and their extension to more advanced topics. Beginning with the axiomatic properties of the real number system, students will study sequences, series, limits, and continuity. The course then develops the concepts of differentiability, Taylor expansions, and power series.

In the second half, the focus shifts to the Riemann integral, uniform convergence, normed vector spaces, and the differential calculus of multivariable functions. Throughout, emphasis is placed on precise definitions, logical reasoning, and the construction of proofs. By the end of the semester, students will have acquired both a solid mastery of fundamental results in analysis and the ability to read, write, and communicate rigorous mathematical arguments, preparing them for further study in pure and applied mathematics.

2. Learning Outcomes

Core Competencies				
Problem Solving	Reasoning	Communication	Attitude and Practice	Problem Identification and Formulation

3. Textbooks and References

- a. Primary Textbook: Robert G. Bartle & Donald R. Sherbert
– *Introduction to Real Analysis*, 4th edition
- b. Reference Books:
 - Stephen Abbott – *Understanding Analysis*, 2nd edition
 - Gerald B. Folland – *Advanced Calculus*, 2nd edition
 - Tom M. Apostol – *Mathematical Analysis*, 2nd edition

4. Teaching Methods

- a. Lectures on rigorous concepts, definitions, and proofs
- b. Student presentations and discussions on selected problems
- c. Homework assignments and exams for evaluation

5. Evaluation Criteria

a. Components

Assignments & Participation	Midterm	Final Exam	Attendance & Attitude	Total
30	100	140	30	300

b. Methods

1. **Exams (Midterm, Final):** 170 minutes each, closed book, proof/essay format. Partial credit given. No calculators or electronic devices allowed. Students without a basis score will be required to take a Make-up Examination.
2. **Assignments:** About 4 sets. Deadlines announced separately. Late submission penalty: 20% deduction per day (24h). After 5 days (120h), not graded (0 points). Server timestamp determines submission time.
3. **Presentations/Participation:** Evaluated based on presentations, discussions, and engagement.
4. **Attendance and Grading:** Tardiness: -1 point; Absence: -3 points. Grading is based on absolute standards, with possible small adjustments depending on exam difficulty.

6. Grading Policy

- Grades: A(+ / 0 / -), B(+ / 0 / -), C(+ / 0 / -), D(+ / 0 / -), F.
- Absolute grading standard will be applied.

7. Weekly Schedule

Week (Period)	Topics and Content	Remarks
Week 1 Aug. 18–22	Sequences of real numbers, limits, algebra of limits Monotone sequences, subsequences, Bolzano–Weierstrass	
Core Competency	Reasoning, Problem Identification and Formulation	
Week 2 Aug. 25–29	Cauchy sequences and completeness Infinite series: definitions, basic properties	
Core Competency	Problem Solving, Reasoning	
Week 3 Sept. 1–5	Convergence tests (comparison, ratio, root, alternating) <i>Transition to Continuity</i> : limits of functions, basic continuity	
Core Competency	Problem Solving, Communication	
Week 4 Sept. 8–12	Properties of continuous functions; continuity on closed intervals Intermediate Value Theorem, Extreme Value Theorem <i>Start Differentiability</i> : definition as a linear approximation	
Core Competency	Reasoning, Problem Identification and Formulation	
Week 5 Sept. 15–19	Basic Properties of Differentiation Mean Value Theorem and applications	
Core Competency	Problem Solving, Reasoning, Attitude and Practice	
Week 6 Sept. 22–26	Higher derivatives, Taylor’s theorem and error estimates <i>Transition to Power Series</i> : radius of convergence	
Core Competency	Problem Identification and Formulation, Communication	
Week 7 Sept. 29–Oct. 3	Exponential and trigonometric functions Termwise operations	National Foundation Day (Oct. 3)
Core Competency	Problem Solving, Reasoning	
Week 8 Oct. 6–10	No class (Chuseok, Hangeul Day, School Holiday)	

Week (Period)	Topics and Content	Remarks
Week 9 Oct. 13–17	Midterm Exam	Oct. 17: Regular class
Week 10 Oct. 20–24	Riemann integral: definition, integrability criteria Basic properties of the integral	
Core Competency	Problem Solving, Reasoning	
Week 11 Oct. 27–31	Fundamental Theorem of Calculus Elementary consequences of the FTC, Improper integrals; integral test	
Core Competency	Problem Identification and Formulation, Communication	
Week 12 Nov. 3–7	Sequences and series of functions; pointwise vs uniform convergence Uniform convergence and interchange of limits	
Core Competency	Problem Solving, Reasoning	
Week 13 Nov. 10–14	Normed vector spaces: norms, equivalent norms, completeness Open/closed sets, limit points, compactness (Heine–Borel in $\mathbb{R}^n$)	
Core Competency	Reasoning, Problem Identification and Formulation	
Week 14 Nov. 17–21	Equivalence of norms in finite dimensional spaces Differentiation $\mathbb{R}^n \rightarrow \mathbb{R}^m$: definition via linear maps Operator norm and continuity of linear maps; Chain Rule	
Core Competency	Problem Solving, Communication	
Week 15 Nov. 24–28	Second derivatives, symmetry of mixed partials Multivariable Taylor theorem and applications	
Core Competency	Problem Solving, Reasoning	
Week 16 Dec. 1–5		Makeup Week
Week 17 Dec. 8–12	Final Exam	

※ The course schedule is subject to change according to the academic calendar.